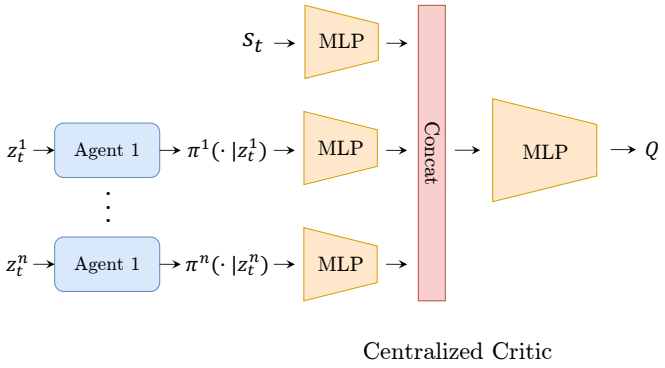

Figure A2: Architecture sketch for LICA with an MLP critic (C_{MLP}, Section 3.2).

Figure A3: Example agent formation by LICA (red units on the left) in StarCraft II `5m_vs_6m` where Marines with high health move forward while the dying Marine is pulled back. Green squares on the floor are agent `move` targets.

cooperative strategies require the agents to reason about higher level battle dynamics despite acting on their local observations in a decentralized manner.

On the other hand, `MMM2` (1 Medivac + 2 Marauders + 7 Marines vs 1 Medivac + 3 Marauders + 8 Marines) features units with different functionalities (Medivac is a healer unit), health (Marauder is more endurable), attack damages, and attack speeds (Marauders has a higher damage than Marines, but their attacks are slower). This allows more complex cooperative strategies between the agents. From the demo video, we first observe that LICA agents learned to prioritize killing enemy Marines because they are brittle but have higher DPS (damage per second) overall. In particular, the agents learned that the enemy Marines attacking the ally Medivac should be targeted first. We also observe that LICA agents learned to pull Marauders forward despite their longer shooting range compared to Marines; this is because Marauders are more durable and can afford to divert possible enemy attention and take more incoming damage. Moreover, unlike ally Marines, the two Marauders did not focus-fire the enemy Marines but instead chose their own targets with high health; we believe that this is to prevent *overkill* (dealing more damage than necessary to kill a unit) because their slow projectile and attack speeds imply that their attacks are harder to plan and easier to miss on enemy units that are already being focus-fired. Furthermore, we observe that the ally Medivac learned to switch between multiple ally units and prioritize units with low health. In particular, it learned to sacrifice (by not healing) a dying Marauder unit to focus on other units that could still be saved; this allows the battle to be won with only one Marauder killed while all other units remain.